Ally Stauffer

+1 (925) 357-7542 | astauf02@calpoly.edu | linkedin.com/in/allystauffer | github.com/allysonstauffer | allystauffer.vercel.app

EDUCATION

Bachelor of Science, Computer Science | 3.64 GPA *Expected May 2028*

California Polytechnic State University, San Luis Obispo *San Luis Obispo, CA*

Relevant Coursework: Data Structures (Java), Object-Oriented Programming & Design (Java), Algorithms, Database Management Systems, Systems Programming, Data Science, Enterprise Analytics, Linear Analysis, Discrete Structures

WORK EXPERIENCE

Software Engineer *Aug 2026 – Present*

Cal Poly Noyce School of Applied Computing *San Luis Obispo, CA*

- Selected as the **primary student Software Engineer** developing **infrastructure monitoring and data visualization** for Cal Poly's **$3M NVIDIA-powered AI Factory**, spanning four DGX B200 systems, compute, networking, storage, and power infrastructure.
- Designing **Python telemetry pipelines using NVIDIA Run APIs** to extract and consolidate running/pending workloads, GPU resource requests and allocations, utilization, and capacity metrics.
- Collaborating with infrastructure engineers and university leadership to define **data requirements, telemetry schemas, and stakeholder-facing dashboards** supporting workload visibility, capacity planning, and resource decisions.

Software Engineering Intern *Jun 2026 – Aug 2026*

Kestrix *London, England, UK*

- Built and deployed a **Python → BigQuery SQL → Dataform → Looker Studio pipeline** consolidating fragmented drone-imagery validation signals into **hourly, property-level reporting**, identifying incomplete coverage in hours rather than days.
- Engineered **Python scoring and remediation logic** evaluating RGB/thermal image completeness, quality, coverage, and reconstruction status against a **99% completeness threshold** to recommend re-uploads or targeted/full-zone reflights.
- Developed **47 automated pytest cases across 11 fixtures**, validating scoring, remediation decisions, environment loading, and edge cases before reproducing verified Python logic in **production BigQuery SQL**.
- Investigated **7 PostgreSQL/BigQuery entities**, built **SQLAlchemy/Folium geospatial visualizations** of planned vs. collected imagery, and shipped a **filterable, exportable Looker Studio work queue** for Drone Operations.

Next-Gen Innovator *Sep 2025 – Jun 2026*

Apple Next Gen Innovators (via Cal Poly) *Remote*

- Selected as **one of ~10 Cal Poly students** for Apple's inaugural technical mentorship program spanning **22 universities nationwide**, developing an independent AI-image detection reliability project.
- Built **TrustFrame, a detector-agnostic Python CLI evaluation framework** transforming prediction CSVs into reproducible JSON reports with precision/recall, false-negative rates, confusion matrices, and **10-bin expected calibration error**.

Undergraduate Research Assistant *Sep 2024 – Jun 2026*

Cal Poly Computer Science and Software Engineering Department *San Luis Obispo, CA*

- Analyzed participant interactions in a **controlled AI-labeled vs. human-labeled tutoring study**, examining how perceived AI involvement influences user trust, communication, and problem-solving.
- Developed a **React/TypeScript experimental interface** for structured participant data collection; **designed and conducted an Instagram engagement study** examining responses to manipulated content ratings and subsequent user reflection.

Software Developer, Mustang Maps *Oct 2025 – Jun 2026*

CodeBox *San Luis Obispo, CA*

- Shipped features for a **production React Native navigation app with 161+ downloads**, contributing to A*-based routing across a **GeoJSON graph of ~5.7K nodes, 6.3K edges, and 98 km of paths** on an 8-person team.
- Developed a **MapLibre classroom-wayfinding interface** spanning **92 floor-plan zones, 45 buildings, and 4,271 classrooms**; implemented GO/Exit navigation-mode interactions for turn-by-turn routing.

Student Communications Lead *Feb 2026 – Present*

Cal Poly Noyce School of Applied Computing *San Luis Obispo, CA*

- Lead Noyce's **cross-platform communications team**, managing technical storytelling, digital campaigns, and student outreach across university channels; an incoming-student campaign generated **500+ new Instagram followers**.
- Collaborate with **university leadership** to communicate AI research and computing initiatives; selected as **one of two student speakers** to address **400+ incoming students**.

TECHNICAL SKILLS

Languages | Python, SQL, TypeScript, JavaScript, Java, C/C++, HTML/CSS

Data Engineering & Analytics | BigQuery, Dataform, PostgreSQL, Pandas, SQLAlchemy, ETL Pipelines, Looker Studio

Frameworks & APIs | React, React Native, Next.js, Node.js/Express, REST APIs, JSON

Testing & Tools | pytest, Git, Folium, Excel, Claude Code, Cursor